

Motion Planning and Incremental Nonlinear Dynamic Inversion for Aggressive Quadrotor Trajectory Tracking on the Crazyflie 2.x Platform Family

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Abstract

This report describes the design, implementation, and experimental evaluation of a full autonomous-flight stack: minimum-snap motion planning with automatic time allocation, a differential-flatness-based geometric SE(3) baseline controller, and an Incremental Nonlinear Dynamic Inversion (INDI) attitude controller. All three components were deployed and compared across three quadrotor platforms: a standard Crazyflie 2.1, a thrust-upgraded Crazyflie 2.1, and a Crazyflie 2.1 Brushless (CF21BL). Each platform was commissioned end-to-end (motor identification via RPM, filter tuning, cascaded gain tuning) and evaluated on a shared figure-8 tracking benchmark, plus oval and circle trajectories swept across a range of planner aggressiveness. The main result is a controller-and-platform comparison showing that full INDI reduces XY tracking RMSE by 37–60% relative to the geometric SE(3) baseline. The brushless platform’s higher motor bandwidth ($\tau \approx 44$ ms vs. 71 ms for the thrust-upgraded platform) allows more aggressive attitude gains and gives the best overall tracking performance (2.45 cm mean RMSE, $n = 13$), but also brings a resonant ≈ 7.2 Hz attitude oscillation that the thrust-upgraded platform’s lower bandwidth avoids. We trace the root cause of this oscillation (closed-loop resonance at $\omega_n = \sqrt{K_R}$, with actuator lag as the dominant and EKF attitude-estimate lag as a secondary phase-loss term) and design a targeted band-reject (notch) filter as a fix; it is implemented and build-verified but has not yet been flight-tested. The remaining sections cover related work, the planning and control methods used, the experimental setup, the results, a discussion of the cross-platform differences, and the project’s open limitations.

1 Introduction

Tracking aggressive trajectories accurately relies on two complementary techniques. The first is **differential-flatness-based motion planning**, which produces dynamically feasible, minimum-effort trajectories together with closed-form full-state references derived directly from the position trajectory and its derivatives. The second is **Incremental Nonlinear Dynamic Inversion (INDI)**, a control technique that replaces an explicit dynamics model with a real-time measurement of the vehicle’s actual angular acceleration, so the attitude loop stays robust to unmodeled effects and platform-specific uncertainty.

This project implements both. A minimum-snap planning pipeline with automatic time allocation (Richter et al. [3]) generates the trajectories, and full INDI attitude control (Tal & Karaman [1]), including the RPM-feedback-based torque estimation it requires, tracks them on real Crazyflie hardware. We commissioned the full stack end to end (motor identification via RPM, filter tuning, cascaded gain tuning) on three physical platforms: a standard, a thrust-upgraded, and a brushless CF21BL Crazyflie 2.x. This lets us see how platform-specific actuator dynamics

limit the achievable control bandwidth and tracking accuracy. Each platform is benchmarked against a common geometric SE(3) baseline and pushed across a planner-aggressiveness sweep (k_t) to find its practical ceiling.

Two originally planned phases were dropped once a data-confirmed attitude resonance on the highest-bandwidth platform became the project’s main open question (Section 6.2): commissioning a third platform, and a dedicated accel-pinned aerobatics phase. Part of that aerobatics work, a fully inverted loop, was still completed as bonus work once time allowed (Section 7). It uses a separate, explicit-attitude planning approach that is not one of the two automatic-time-allocation planning modes used for every other experiment in this report (Section 3.1).

2 Related Work

A quadrotor’s translational dynamics are differentially flat [2]: the full 6-DOF state and the control inputs can be written as algebraic functions of the position trajectory $\mathbf{p}(t)$, yaw $\psi(t)$, and a finite number of their derivatives (up to snap for the full state, one derivative further for angular-acceleration feedforward). A smooth polynomial in position/yaw space is therefore enough to derive a complete, dynamically consistent reference without ever planning attitude directly. Our baseline attitude controller follows the geometric SE(3) approach of Lee et al. [4], which computes attitude and rate errors directly on SO(3), avoiding the singularities of an Euler-angle representation, and commands torque model-based from these errors plus a gyroscopic decoupling term. Trajectories are generated with Mellinger and Kumar’s [2] minimum-snap QP formulation, extended by Richter et al. [3] to automate per-segment timing from a single aggressiveness scalar k_t . The INDI attitude law (Tal & Karaman [1]) replaces the geometric law’s model-based feedback with an incremental update: the next torque command is the currently applied torque plus a correction proportional to the gap between the desired and the measured angular acceleration. Because any unmodeled effect shows up directly in that measurement, INDI stays robust to model mismatch without needing the mismatch to be characterized in advance. This property turns out to matter a great deal for the main practical challenge of this project (Section 6.2).

3 Methods

3.1 Motion planning

All trajectories in this project are generated by one of two automatic-time-allocation planning modes. Both are built on the same minimum-snap QP machinery (`src/planning/richter.rs`) and both are driven by a single scalar aggressiveness parameter k_t rather than manually chosen segment durations:

- **Mode 1** (`RichterTrajectory::plan()`) uses Richter et al.’s [3] time-allocation formula,

$$T_i = \max\left(\frac{\text{dist}_i}{v_{\text{avg}}(k_t)}, T_{\min}\right), \quad v_{\text{avg}} = 0.5 + 1.5\sqrt{k_t}, \quad T_{\min} = 0.15 \text{ s}$$

below which the QP’s $1/T^7$ segment weighting becomes numerically degenerate. We use zero redistribution iterations by default, favoring robustness and predictability over the marginal timing refinement the full paper formulation offers.

- **Mode 3** (`RichterTrajectory::plan_paper()`) uses the same time-allocation formula but adds the full eight gradient-descent redistribution iterations specified in Richter et al. [3], which push segment time toward the highest-snap portions of the trajectory for more balanced actuator demand across the maneuver.

Both modes share the same differential-flatness recovery, following the Faessler et al. [5] formulation (implemented in `src/planning/flatness.rs`). Given the position polynomial’s derivatives at any query time t (velocity, acceleration, jerk, snap) plus yaw and its derivatives, the full rigid-body reference is recovered in closed form:

$$\mathbf{a}_g = \ddot{\mathbf{p}} + g\hat{\mathbf{z}} \quad (1)$$

$$\mathbf{x}_b = \text{normalize}(\mathbf{y}_c \times \mathbf{a}_g), \quad \mathbf{y}_b = \text{normalize}(\mathbf{a}_g \times \mathbf{x}_b), \quad \mathbf{z}_b = \mathbf{x}_b \times \mathbf{y}_b \quad (2)$$

$$c = \mathbf{z}_b \cdot \mathbf{a}_g, \quad T = m c \quad (3)$$

$$\boldsymbol{\omega} = f(\mathbf{j}, \mathbf{x}_b, \mathbf{y}_b, c) \quad (\text{from jerk } \mathbf{j} = \ddot{\mathbf{p}}) \quad (4)$$

$$\dot{\boldsymbol{\omega}} = f(\mathbf{s}, \boldsymbol{\omega}, \mathbf{x}_b, \mathbf{y}_b, c) \quad (\text{from snap } \mathbf{s} = \mathbf{p}^{(4)}) \quad (5)$$

$$\boldsymbol{\tau} = J\dot{\boldsymbol{\omega}} - (J\boldsymbol{\omega}) \times \boldsymbol{\omega} \quad (6)$$

Here $\mathbf{x}_c = [\cos \psi, \sin \psi, 0]$ and $\mathbf{y}_c = [-\sin \psi, \cos \psi, 0]$ define the yaw-heading intermediate frame used to disambiguate the body- x /body- y axes from the thrust-direction-only $\mathbf{z}_b$. This closed-form map lets the entire trajectory library used in this project (figure-8, circle, oval, corner, helix, tilted oval) be authored purely as position/yaw waypoints. Attitude, angular rate, angular acceleration, and thrust are all derived from that, never planned separately.

3.2 Geometric SE(3) baseline control

The geometric controller [4] is the same control law used across all three platforms and is INDI’s structural baseline throughout this report. The formal, validated geometric-vs-INDI comparison in Section 5.2 (Table 1) was carried out on the standard platform only; the thrust-upgraded and brushless platforms are compared to each other and to that standard-platform reference through their own INDI results, not through their own separate geometric campaigns. Attitude error is computed on SO(3):

$$\mathbf{e}_R = \frac{1}{2} \text{vee} \left(R_d^\top R - R^\top R_d \right)$$

and torque is commanded directly from a model-based law:

$$\boldsymbol{\tau} = -K_R \mathbf{e}_R - K_W \mathbf{e}_\omega + \boldsymbol{\omega} \times J\boldsymbol{\omega}$$

with K_R and K_W tuned per platform (the position loop uses K_P/K_V on position/velocity error with the same cascade structure). Because this law is purely model-based, its accuracy depends on how closely J (inertia) and the thrust/torque allocation model match the real airframe.

3.3 INDI attitude control

The INDI attitude law, following Tal & Karaman [1] directly, replaces the geometric law’s model-based torque computation with four coupled equations.

3.3.1 (1) Feedforward

The desired angular acceleration $\boldsymbol{\alpha}_{\text{des}}$ is computed from the trajectory’s snap via differential flatness (Tal & Karaman [1], Eq. 15). This is the only place the desired trajectory’s higher derivatives feed the attitude loop; everything else below is feedback.

3.3.2 (2) Reference blend

$$\boldsymbol{\alpha}_{\text{ref}} = \boldsymbol{\alpha}_{\text{des}} - K_R \mathbf{e}_R - K_W \mathbf{e}_\omega \quad (\text{Eq. 28})$$

This has the same form as the geometric law’s gains, but α_{ref} is a desired acceleration, not a torque. K_R has units $1/\text{s}^2$ (closed-loop bandwidth $\omega_n = \sqrt{K_R}$) and K_W has units $1/\text{s}$ (damping ratio $\zeta = K_W/(2\omega_n)$).

3.3.3 (3) Measured angular acceleration

No sensor measures angular acceleration directly, so α_{meas} is derived from the gyroscope through a two-stage chain, matching the order specified by Tal & Karaman [1] Sec. III-D and the stock Crazyflie firmware: the raw body rate is first low-pass filtered, then differentiated.

$$\omega_{\text{filt}}[k] = \text{BW}(\omega_{\text{raw}}[k]), \quad \alpha_{\text{meas}}[k] = \frac{\omega_{\text{filt}}[k] - \omega_{\text{filt}}[k-1]}{\Delta t}$$

The same Butterworth filter is applied identically to α_{ref} , which keeps the reference and measured legs phase-matched before their difference is taken in the increment law below.

3.3.4 (4) Incremental torque law

$$\delta\tau = J(\alpha_{\text{ref}} - \alpha_{\text{meas}}) \quad (7)$$

$$\tau_{\text{new}} = \tau_{\text{current}} + \delta\tau \quad (\text{Eq. 31}) \quad (8)$$

This is the core INDI mechanism: a small correction to whatever torque is currently applied, using measured, not modeled, angular acceleration. τ_{current} is estimated from the physical system, per-motor RPM converted to thrust via a bench-identified motor constant k_T and then mapped to body torque via the known arm geometry, rather than from the previous commanded torque, which would not account for real motor response lag. This is the key structural difference from the geometric law: INDI does not need an accurate model of thrust-to-RPM response, aerodynamic drag, or other unmodeled disturbances, because any such effect already shows up in α_{meas} and is rejected by the increment on the next cycle.

3.4 An added filter stage: targeting a diagnosed narrowband resonance

Brushless-platform commissioning identified a narrowband attitude oscillation at ≈ 7.22 Hz (Section 6.2 gives the full root-cause analysis). We confirmed by 500 Hz SD-card measurement that the Stage-1 Butterworth filter (cutoff 60–70 Hz) passes this frequency essentially unattenuated ($|\alpha_{\text{meas}}|/|\alpha_{\text{raw}}| = 1.000$), and that lowering the cutoff into that band degrades tracking substantially, since a broad low-pass trades off phase margin below its cutoff, not just at the resonant band. We designed a targeted fix instead: an optional, runtime-tunable second-stage band-reject (notch) filter, an RBJ Audio-EQ-Cookbook band-reject biquad ($Q = f_0/\text{bw}$, default center 7.2 Hz, bandwidth 5 Hz, giving a 4.7–9.7 Hz attenuated band). When enabled it is applied identically to α_{meas} , α_{ref} , and τ_{current} , so no term of the INDI sum ends up phase-mismatched, mirroring the Stage-1 Butterworth’s own reasoning. It is disabled by default (`notch_en=0`, verified byte-identical to prior behavior when off) and is implemented and build-verified but not yet flight-tested (Section 7).

4 Experimental Setup

We commissioned three platforms: a **standard** Crazyflie 2.1 (brushed 0716 motors, 36.4 g, $J_{xx} = J_{yy} = 1.657 \times 10^{-5} \text{ kg m}^2$, used as the reference platform), a **thrust-upgraded** Crazyflie 2.1 (brushed upgrade kit, 38.6 g, similar inertia, higher peak thrust per motor at 0.18 N vs. 0.13 N, but a heavier rotor), and a **brushless** CF21BL (41.0 g, $\text{diag}(2.40, 2.40, 3.24) \times 10^{-5} \text{ kg m}^2$ [6], the highest motor bandwidth of the three, with DShot bidirectional telemetry). Full physical

parameters are listed in Appendix B (Table 2). Every platform was bench-identified for its per-motor thrust constant k_T (thrust stand, RPM sweep, $k_T = \text{thrust}/\text{RPM}^2$) before any INDI flight.

Trajectories are generated offline in Rust (`export_poly4d`), converted to an 8-coefficient Poly4D representation, and uploaded via the CrazySwarm2 (CS2) ROS 2 framework. Position is estimated from OptiTrack and fed to the onboard EKF. The onboard high-level commander evaluates the uploaded polynomial at 1 kHz and feeds the geometric or INDI attitude controller, running at 500 Hz, through the standard Crazyflie firmware loop. This gives globally drift-free pose for repeatable long-duration, high-speed flights.

The primary metric throughout is **phase-aligned XY position RMSE**: the flown path is time-shifted to maximize cross-correlation with the planned path before computing RMSE, which corrects for any constant commanded-vs-flown lag. Secondary metrics are roll/pitch tracking error and, for stress-testing, per-axis gyro standard deviation as a proxy for attitude oscillation severity. We classify a flight as a **crash** (used throughout Section 5.3 and Appendix D) when logged altitude dives well below the commanded trajectory, which is physically impossible under controlled flight, and this is corroborated by a simultaneous jump in gyro standard deviation. Both signals are unambiguous in the logged telemetry and were cross-checked against each flight’s own recording. All exported trajectories were validated against a physically confirmed usable flight volume ($x: \pm 1.2$ m, $y: \pm 2.5$ m, $z: 0.5$ – 1.8 m) by densely sampling every trajectory across the full k_t range flown and checking worst-case bounding boxes before any flight.

The core benchmark for every cross-platform, cross-controller comparison is a **figure-8** at $k_t = 0.05$ (lap ≈ 6.9 s). Extended aggressiveness sweeps additionally use **oval** and **circle** shapes at increasing k_t to find each platform’s practical ceiling.

5 Results

5.1 Per-platform tuning summary

Full locked-configuration tables (all gains, filters, and bench-identified thrust constants) are in Appendix B (Tables 3–5).

Standard CF2.1. A filter cutoff sweep found $fc_bw = 70$ Hz optimal (3.1 cm RMSE). The final configuration is $K_R = 1050$, $K_W = 87$ ($\omega_n \approx 32.4$ rad/s), $K_{P,xy} = 40$, $K_{V,xy} = 3.3$. Over 10 flights, INDI averaged **3.87 cm** RMSE against 6.14 cm for the geometric baseline, a 37% reduction (Figure 1, Appendix C).

Thrust-upgraded CF2.1. The attitude gain ladder found a hard ceiling at $K_R = 603$ ($\omega_n \approx 24.6$ rad/s), well below the standard platform’s 1050, despite the upgrade kit’s higher peak thrust (0.18 N vs. 0.13 N per motor). The final configuration is $K_R = 603$, $K_W = 90$ ($\zeta \approx 1.83$, $\tau_{slow} = 137$ ms), $K_{P,xy} = 40$, $K_{V,xy} = 8$. An 11-flight validation gave **3.72 cm** mean RMSE (std 0.21), close to the standard platform despite the lower achievable K_R .

Brushless CF21BL. This platform saw the most extensive tuning campaign of the three. A coupled gain ladder (K_R , K_W , K_P , K_V scaled together) improved RMSE monotonically up to a hover-noise ceiling at $K_R \approx 2400$ – 2600 . Decoupled sweeps then showed that pulling K_W and K_V down independently of that ratio improved both RMSE and attitude error at the same time, a real decoupling win rather than a tradeoff, because K_W injects raw gyro noise directly and linearly through $K_W \mathbf{e}_\omega$. The final configuration is $K_R = 2400$, $K_W = 170$ ($\zeta \approx 1.74$), $K_{P,xy} = 64$, $K_{V,xy} = 5$. A 12-flight validation gave **2.48 cm** mean RMSE (std 0.14), the best of any platform in this project.

Configuration	n	Mean XY RMSE	Std	vs. Standard Geometric
Standard Geometric (baseline)	10	6.14 cm	0.52	—
Standard INDI	10	3.87 cm	0.24	−37%
Upgraded INDI	11	3.72 cm	0.21	−39%
Brushless INDI (fc_bw=60)	12	2.48 cm	0.13	−60%
Brushless INDI (fc_bw=70)	13	2.45 cm	0.12	−60%

Table 1: Five-way headline comparison, figure-8 trajectory, $k_t = 0.05$.

5.2 Headline cross-platform/cross-controller comparison

The fc_bw = 60 vs. 70 Hz brushless columns are otherwise identical configurations, so they isolate the filter cutoff’s effect: the two means (2.48 ± 0.14 vs. 2.45 ± 0.12 cm) fall well within one combined standard deviation of each other across the 20–100 Hz range tested (we did not run a formal significance test). This is consistent with this platform’s tracking ceiling being set by attitude gain, and the resonance it induces (Section 6.2), rather than filter tuning. A bar-chart visualization (Figure 2) and the full per-flight breakdown of the best brushless flight (flown path, 3D attitude, INDI torque, and RPM balance) are reproduced at full size in Appendix C (Figures 3–6).

5.3 Extended aggressiveness sweeps (upgraded vs. brushless)

Full per- k_t data (XY/Z RMSE, roll/pitch/yaw error, max speed) for both sweeps is in Appendix D (Tables 6–7).

Oval. Both platforms flew cleanly up to $k_t \leq 0.3$ –0.4. The upgraded platform crashed consistently at $k_t = 0.5$ –0.6 (3/3 attempts), while brushless stayed clean through $k_t = 0.5$ and crashed only at $k_t = 0.7$. Brushless’s working ceiling is roughly $k_t \approx 0.5$ against upgraded’s $k_t \approx 0.3$ –0.4. This tracks the bandwidth gap between them ($\omega_n \approx 7.8$ Hz for brushless vs. ≈ 3.9 Hz for upgraded), which gives brushless more control authority to correct through sharp direction changes, at the cost of sitting closer to its own resonant regime.

Circle. This shape was newly swept on both platforms in this project; previously only $k_t = 0.05$ had been flown. Both platforms show the same pattern: a clean-flight region followed by a marginal zone where crash and clean outcomes mix at the same k_t rather than switching at a sharp threshold. Upgraded stays clean to ≈ 0.6 , brushless to ≈ 0.5 , and both show failures by $k_t = 0.8$ –1.0. Cross-platform RMSE at matched k_t is tied at $k_t = 0.05$ (1.7 vs. 1.9 cm) but favors brushless at $k_t = 0.3$ (1.4 vs. 2.2 cm, about 36% lower), so which platform tracks better depends on k_t rather than being a fixed ranking.

Tracking-overview plots for both representative points (oval $k_t = 0.4$, circle $k_t = 0.7$) are reproduced at full size in Appendix C (Figures 7–8).

6 Discussion

6.1 Why the platforms rank the way they do

Considering position RMSE together with attitude tracking error gives a clearer ranking than RMSE alone. In the best-flight breakdown (Appendix C, Figure 3), roll error runs 7.2° (standard) $\rightarrow 5.3^\circ$ (upgraded) $\rightarrow 4.9^\circ$ (brushless), and pitch error $5.1^\circ \rightarrow 3.4^\circ \rightarrow 3.2^\circ$ in the same order, with position RMSE following the same order in that flight (3.6, 3.2, 2.2 cm). So on both metrics, brushless tracks best, thrust-upgraded second, standard third, even though the aggregate n -flight RMSE in Table 1 shows standard and upgraded close enough (3.87 vs. 3.72 cm) to fall within one combined standard deviation of each other.

Brushless’s clear lead is well explained by achievable INDI attitude bandwidth, which is set by motor response speed rather than peak thrust. Its bench-measured actuator time constant ($\tau \approx 44$ ms) is faster than the thrust-upgraded platform’s ($\tau \approx 71$ ms), and because INDI’s stability margin depends on how quickly the motor can realize a commanded change in angular acceleration, a faster actuator supports a higher stable K_R (2400 vs. 603) and therefore a higher closed-loop attitude bandwidth. This is a hardware-set advantage rather than a tuning outcome. Brushless leads even with the ≈ 7.22 Hz resonance below still unresolved; fixing it should widen this lead, not shrink it.

The smaller gap between standard and upgraded does not fit the same explanation as cleanly: the standard platform reaches a higher stable K_R (1050 vs. 603) yet tracks slightly worse on every metric above. We do not have a confirmed explanation for this reversal; it may reflect differences elsewhere in the position or attitude loop rather than actuator bandwidth, and we note it here rather than force a tidy story that the data does not fully support.

6.2 The brushless attitude resonance: what we observed and our working hypothesis

Two things are directly observed, not inferred. First, the brushless platform’s raw angular-acceleration signal (α_{raw}) shows a narrowband spectral peak at ≈ 7.22 Hz, measured by 500 Hz SD-card logging. The existing Stage-1 Butterworth filter does essentially nothing to it ($|\text{filtered}|/|\text{raw}| = 1.000$), since its cutoff (60–70 Hz) sits well above this band by design. Second, commanded torque on brushless is markedly larger than on the other two platforms across all three axes in the same flight comparison (Appendix C, Figure 5): τ_z RMS is 2.17 vs. 0.19 (standard) and 0.14 (upgraded) N·m, roughly an order of magnitude higher.

Beyond these two observations, what follows is our working hypothesis for the mechanism, not a confirmed cause. Our leading candidate is a feedback resonance: $\omega_n = \sqrt{K_R}/2\pi = 7.80$ Hz at $K_R = 2400$ is close to the measured 7.22 Hz peak (within 8%), consistent with the closed-loop attitude bandwidth sitting near its own resonant frequency and sustaining a self-oscillation there, per the classical Barkhausen/Nyquist condition. Converting the bench-measured actuator lag (31.6–33.6 ms) to phase at 7.22 Hz gives $\approx 82^\circ$, about four times the $\approx 20^\circ$ from EKF attitude-estimate lag, so if this picture is right, actuator lag is the larger of the two measured contributors. Lowering the filter cutoff into the resonant band directly made tracking measurably worse, consistent with a low-pass trading off phase margin across its whole passband to reach one narrow region. None of this proves the mechanism; we have not, for example, varied K_R independently over a wide enough range to isolate it from other explanations. The full 500 Hz diagnostic (hover and oval-maneuver spectra) is in Appendix C (Figures 9–10); it shows the same peak at hover and under real maneuvering load.

Whatever the exact mechanism turns out to be, the two observations alone motivate the fix: a narrowband energy peak sits in the signals that feed the INDI increment, and the existing filter does not touch it. Removing that energy with a notch filter (Section 3.4) should reduce the oscillation it drives, whether or not our resonance explanation for why it sits at 7.22 Hz turns out to be fully correct.

7 Limitations & Future Work

Brushless attitude resonance (open, pending validation). The notch filter is implemented and build-verified (byte-identical to prior behavior when disabled) but not yet flight-tested. We expect a sharp amplitude drop in the notch-filtered signal at ≈ 7 Hz and calmer commanded torque; if the shake is instead dominated by actuator phase lag, this fix may only partially resolve it. Other candidates considered but not pursued, since they need hardware or firmware

iteration beyond this project’s scope: an actuator-lag compensation term in the control law, reducing vibration at the source (prop balancing, mount damping), and, speculatively, motor-protocol-level rate limiting (e.g. DShot slew-rate limiting).

A separate crash that was never root-caused occurred on a different maneuver, a wide teardrop loop, under the plain geometric controller on both platforms at the mildest tested aggressiveness. It was flagged but not investigated further.

Inverted-loop aerobatic maneuver (bonus, beyond core scope). Using a separate explicit-attitude approach outside this report’s core planning methods (Section 3.1), we commissioned a fully inverted vertical loop on the brushless platform: a repeatable inversion (roll 163–178° at the apex, 3/3 flights) and, in the best flight, a real post-inversion recovery. One narrower failure mode remains inside the maneuver’s auto-generated exit ramp. Several fixes were tried and ruled out; the mechanism is well characterized but still unresolved.

Third platform and dedicated aggressive-maneuver phase. Both were descopeed once the resonance investigation became the dominant open question. The inverted-loop work above is a partial, unplanned recovery of the latter.

8 Conclusion

This project implemented minimum-snap motion planning with automatic time allocation, a differential-flatness-derived geometric baseline, and INDI attitude control, then commissioned and evaluated all three on three physically distinct quadrotor platforms. The central result is a 37–60% reduction in figure-8 tracking RMSE from INDI over the geometric baseline, governed almost entirely by each platform’s achievable motor bandwidth rather than by tuning effort. The trajectory library was pushed beyond the core benchmark with oval and circle sweeps, which mapped out real, platform-specific speed ceilings. The project’s most significant open finding, a narrowband ≈ 7.22 Hz attitude resonance on the highest-bandwidth platform, is documented with the observations behind it, a working hypothesis for its mechanism, and a data-driven fix, a notch filter, that is implemented and ready for validation. Together with an unplanned excursion into fully inverted aerobatic flight, the limitations reported here are, in every case, well-characterized open problems rather than unexplained failures.

References

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- [5] M. Faessler, A. Franchi, and D. Scaramuzza, “Differential Flatness of Quadrotor Dynamics Subject to Rotor Drag for Accurate Tracking of High-Speed Trajectories,” IEEE RA-L, 2018.
- [6] A. Busetto et al., “Nonlinear System Identification Nano-drone Benchmark,” 2025 (brushless platform parameters).

A Full INDI and Notch-Filter Equations

This appendix collects the complete equation set summarized in the Methods section, including implementation-level detail (exact filter coefficients, the motor allocation matrix) omitted from the main text for readability.

A.1 Snap-based angular acceleration feedforward

The desired angular acceleration $\boldsymbol{\alpha}_{\text{des}}$ (Tal & Karaman [1], Eq. 15) is computed in closed form from the trajectory's jerk $\mathbf{j} = \ddot{\mathbf{p}}$ and snap $\mathbf{s} = \mathbf{p}^{(4)}$, for constant yaw ($\dot{\psi} = 0$):

$$\dot{c} = \mathbf{b}_3 \cdot \mathbf{j} \quad (9)$$

$$\alpha_x = \frac{2(\mathbf{y}_b \cdot \mathbf{j})(\mathbf{b}_3 \cdot \mathbf{j})}{c^2} - \frac{\mathbf{y}_b \cdot \mathbf{s}}{c} \quad (10)$$

$$\alpha_y = \frac{\mathbf{x}_b \cdot \mathbf{s}}{c} - \frac{2(\mathbf{x}_b \cdot \mathbf{j})(\mathbf{b}_3 \cdot \mathbf{j})}{c^2} \quad (11)$$

$$\alpha_z = 0 \quad (12)$$

where $c = \|\mathbf{a} + g\hat{\mathbf{z}}\|$ and $\mathbf{x}_b, \mathbf{y}_b, \mathbf{b}_3$ are the desired body axes from the flatness recovery (Section 3.1). This is a pure feedforward term computed once per trajectory sample; no filtering is applied to $\boldsymbol{\alpha}_{\text{des}}$ itself, since it is a planned reference signal, not a noisy measurement.

A.2 Reference blend and rate error

$$\mathbf{e}_\omega = \boldsymbol{\omega} - \boldsymbol{\omega}_d \quad (13)$$

$$\boldsymbol{\alpha}_{\text{ref}} = \boldsymbol{\alpha}_{\text{des}} - K_R \mathbf{e}_R - K_W \mathbf{e}_\omega \quad (14)$$

where $\boldsymbol{\omega}_d$ (desired body angular rate) is itself a differential-flatness feedforward term derived from jerk (Tal & Karaman [1], Eq. 14), and $\mathbf{e}_R$ is the SO(3) attitude error defined in Section 3.2.

A.3 Measured angular-acceleration filter chain

Second-order Butterworth (bilinear-transform biquad, cutoff f_c , sample period Δt), applied to the raw body rate $\boldsymbol{\omega}_{\text{raw}}$:

$$y[k] = b(x[k] + 2x[k-1] + x[k-2]) - a_1y[k-1] - a_2y[k-2]$$

with the standard bilinear-transform coefficients b, a_1, a_2 derived from f_c and Δt .

Finite difference, applied to the filtered rate:

$$\boldsymbol{\alpha}_{\text{meas}}[k] = \frac{\boldsymbol{\omega}_{\text{filt}}[k] - \boldsymbol{\omega}_{\text{filt}}[k-1]}{\Delta t}$$

The identical Butterworth filter (same cutoff, independent instances) is applied to $\boldsymbol{\alpha}_{\text{ref}}$ to keep the reference and measured legs of the increment phase-matched before their difference is taken in Eq. A.5 below.

A.4 Optional stage-2 notch (band-reject) filter

The added filter of Section 3.4 is a standard RBJ Audio-EQ-Cookbook band-reject biquad, Direct Form I, parameterized by center frequency f_0 and bandwidth bw (both runtime-tunable):

$$\omega_0 = 2\pi f_0 \Delta t, \quad Q = \frac{f_0}{\text{bw}} \quad (15)$$

$$\alpha = \frac{\sin \omega_0}{2Q}, \quad a_0 = 1 + \alpha \quad (16)$$

$$b_0 = b_2 = \frac{1}{a_0}, \quad b_1 = a_1 = \frac{-2 \cos \omega_0}{a_0}, \quad a_2 = \frac{1 - \alpha}{a_0} \quad (17)$$

$$y[k] = b_0 x[k] + b_1 x[k-1] + b_2 x[k-2] - a_1 y[k-1] - a_2 y[k-2] \quad (18)$$

With default $f_0 = 7.2 \text{ Hz}$, $\text{bw} = 5.0 \text{ Hz}$ ($Q \approx 1.44$), the -3 dB attenuated band is approximately $f_0 \pm \text{bw}/2 \approx 4.7\text{--}9.7 \text{ Hz}$, matching the diagnosed shake band. When enabled, this same filter (independent instances, identical coefficients) is applied to α_{meas} , α_{ref} , and the increment-base term τ_{current} (Eq. A.7 below), so that all three terms entering the final torque command are attenuated identically at the resonant band.

A.5 Incremental torque law and RPM-based increment base

$$\delta \tau = J(\alpha_{\text{ref}} - \alpha_{\text{meas}}) \quad (19)$$

$$\tau_{\text{new}} = \tau_{\text{current}} + \delta \tau \quad (20)$$

τ_{current} is estimated from measured per-motor RPM rather than the previous commanded torque:

$$F_i = k_{T,i} \cdot \text{RPM}_i^2, \quad i = 1, \dots, 4 \quad (21)$$

$$\tau_x = L(F_3 + F_4 - F_1 - F_2) \quad (22)$$

$$\tau_y = L(F_2 + F_3 - F_1 - F_4) \quad (23)$$

$$\tau_z = \kappa(F_2 + F_4 - F_1 - F_3) \quad (24)$$

where L is the motor arm length, κ is the drag-to-thrust torque ratio (motor/propeller dependent, bench/manufacture-identified per platform), and $k_{T,i}$ is the per-motor bench-identified thrust constant. This is the inverse of the standard quadrotor force/torque allocation map: rather than assuming a model of how commanded torque produces motor speeds, the actually-realized motor speeds are read back and converted directly to the torque currently being applied — the mechanism that makes the INDI increment $\delta \tau$ correct for whatever gap exists between the model and reality on every control cycle.

A.6 Differential-flatness torque feedforward (planning side)

For completeness, the planning-side feedforward torque used to derive a dynamically consistent reference trajectory (Section 3.1) is the standard rigid-body inverse dynamics equation with diagonal inertia:

$$\tau_{\text{ff}} = J\dot{\omega} - (J\omega) \times \omega$$

using the flatness-derived ω , $\dot{\omega}$ from Section 3.1's Eqs. (4)–(5). This is distinct from, and only loosely coupled to, the INDI increment law above: the planning-side torque is a purely model-based reference used to check trajectory feasibility (peak torque/thrust against platform limits), while the control-side torque command (Eq. A.6) is what is actually sent to the motor mixer.

B Full Platform Configurations

Platform	Motor type	AUW mass	Inertia $J_{xx} = J_{yy}$	Notes
Standard CF2.1	Brushed (0716)	36.4 g	$1.657 \times 10^{-5} \text{ kg m}^2$	Reference platform; lightest, lowest peak thrust
Thrust- upgraded CF2.1	Brushed (upgrade kit)	38.6 g	$\approx 1.657 \times 10^{-5} \text{ kg m}^2$	Higher peak thrust (0.18 N/motor vs. 0.13 N), heavier rotor $\rightarrow$ slower motor response
Brushless CF21BL	Brushless	41.0 g	diag(2.40, 2.40, 3.24) $\times 10^{-5} \text{ kg m}^2$	Highest motor bandwidth, DSHOT bidirectional telemetry

Table 2: Platform physical parameters. Brushless inertia from Busetto et al. [6].

Attitude (indi_gains)		Position (pos_gains)	
ctrl_mode	3 (full INDI)	kp_xy	40.0
kr, kr_z	1050.0	kp_z	30.0
kw, kw_z	87.0	kv_xy	3.3
fc_bw	70.0 Hz	kv_z	7.0
mass	0.0364 kg		
kt1..kt4	$1.4811, 1.4815, 1.5319, 1.6119 \times 10^{-10} \text{ N/RPM}^2$		

Table 3: Standard CF2.1: final locked configuration.

Attitude (indi_gains)		Position (pos_gains)	
ctrl_mode	3 (full INDI)	kp_xy	40.0
kr, kr_z	603.0	kp_z	30.0
kw, kw_z	90.0	kv_xy	8.0
fc_bw	70.0 Hz	kv_z	10.0
mass	0.0386 kg		
kt1..kt4	$2.2959, 2.4668, 2.0250, 2.5288 \times 10^{-10} \text{ N/RPM}^2$		

Table 4: Thrust-upgraded CF2.1: final locked configuration.

Attitude (indi_gains)		Position (pos_gains)	
ctrl_mode	3 (full INDI)	kp_xy	64.0
kr, kr_z	2400.0	kp_z	48.0
kw, kw_z	170.0	kv_xy	5.0
fc_bw	70.0 Hz	kv_z	7.0
filt_tau	1 (phase-matched increment base)		
j_scale	1.0		
mass	0.041 kg		
kt1..kt4	$4.1623, 4.0592, 4.1116, 4.0631 \times 10^{-10} \text{ N/RPM}^2$		

Table 5: Brushless CF21BL: final locked configuration.

C Extended Flight-Data Figures

The figures below are each dense multi-subplot composites (or, for the two bar charts, simply too wide to sit comfortably in the main text alongside the numeric tables) generated by the project's flight analysis tooling; they are reproduced here at full page width rather than cropped into the main text.

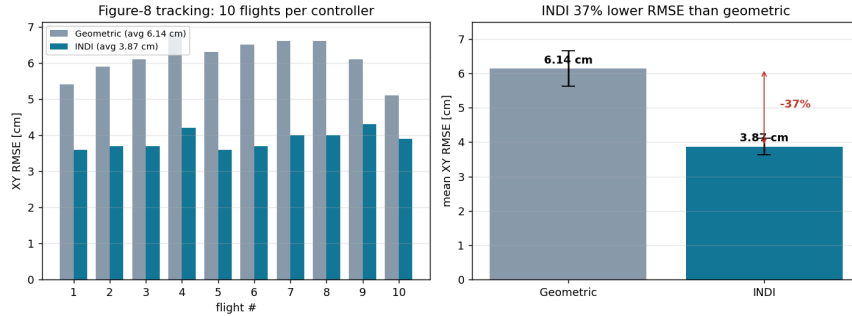


Figure 1: Standard CF2.1: geometric vs. full-INDI, 10 flights each, figure-8 at $k_t = 0.05$.

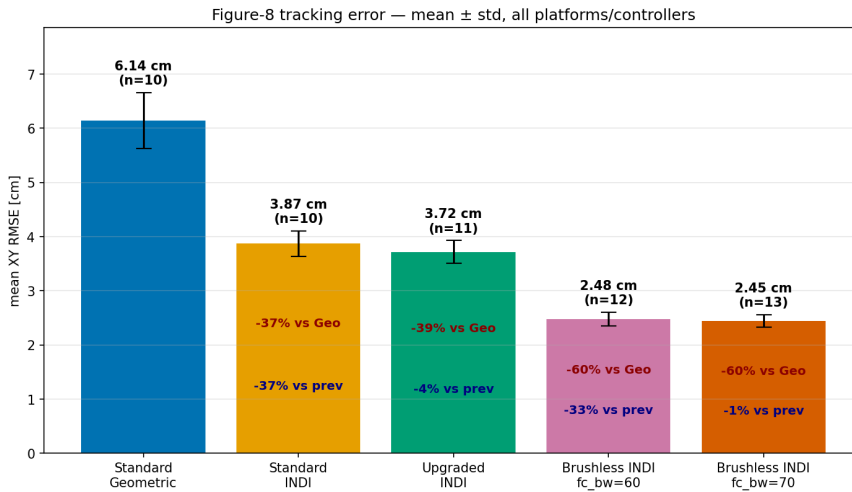


Figure 2: Five-way headline comparison: mean XY RMSE across all platform/controller configurations, figure-8 at $k_t = 0.05$.

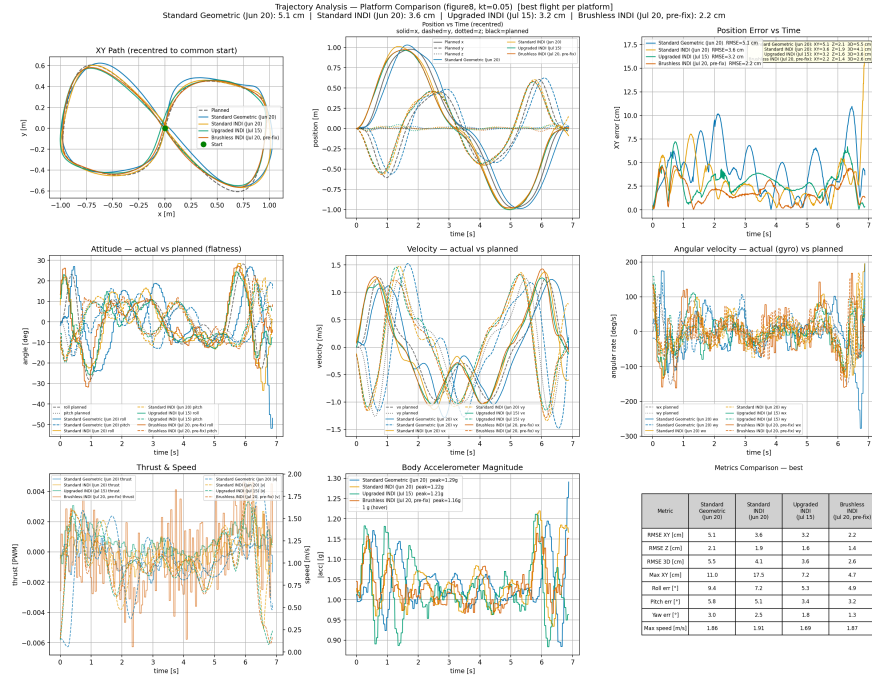


Figure 3: Best flight, brushless platform, figure-8 at $k_t = 0.05$: XY tracking overview.

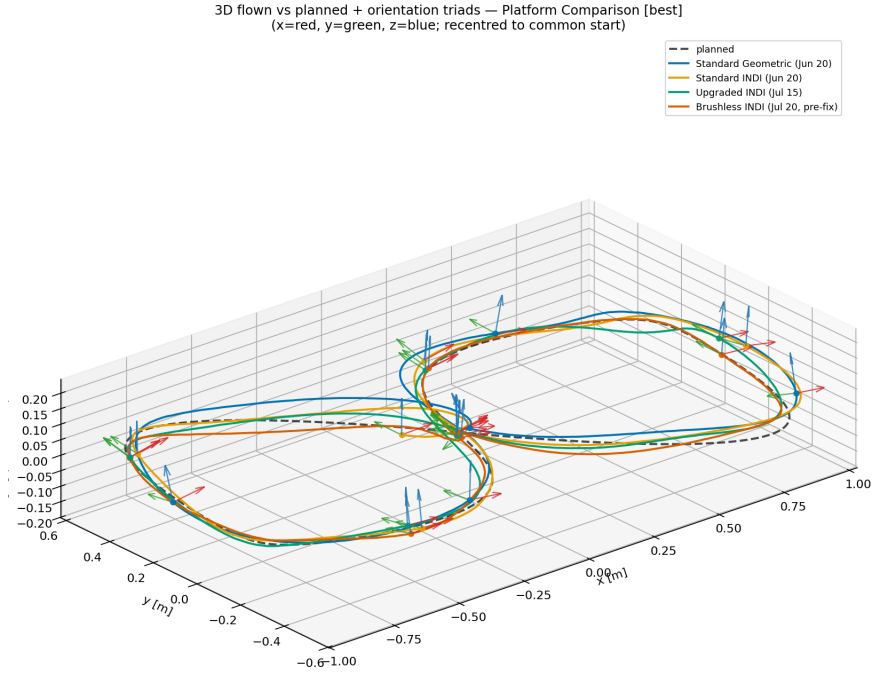


Figure 4: Best flight, brushless platform: 3D orientation.

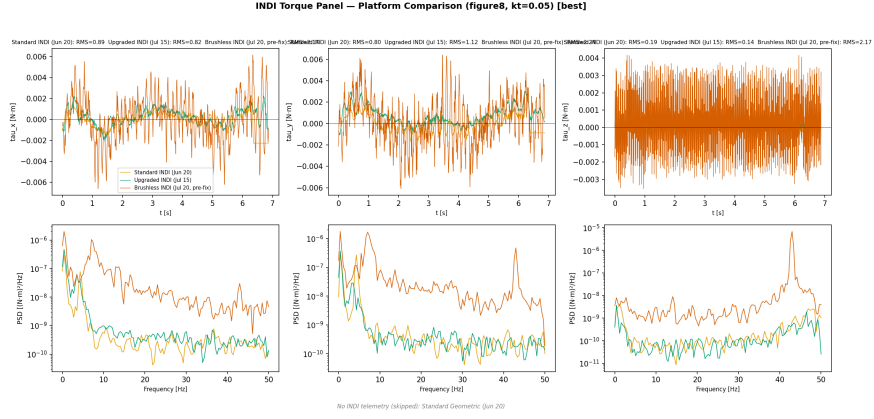


Figure 5: Best flight, brushless platform: INDI torque/angular-acceleration panel.

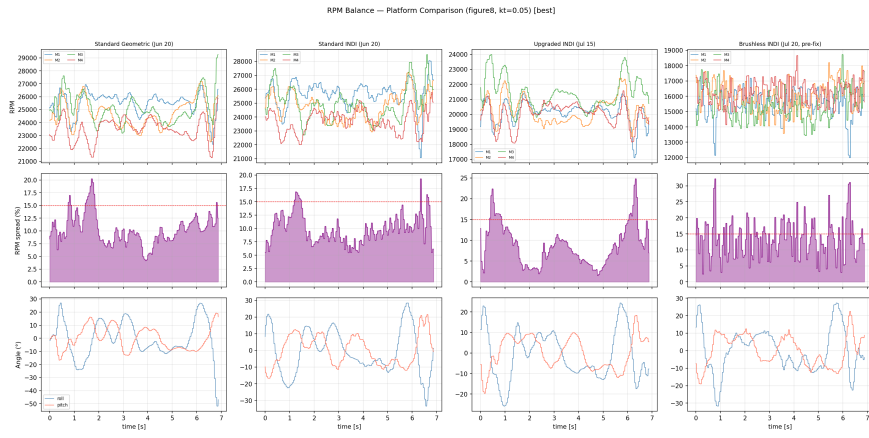


Figure 6: Best flight, brushless platform: per-motor RPM balance.

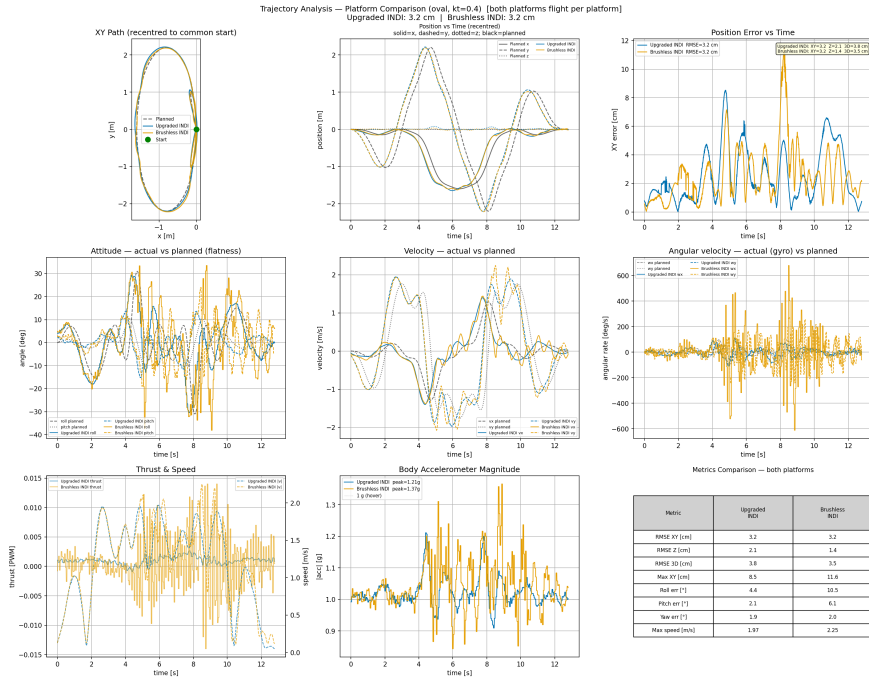


Figure 7: Upgraded vs. brushless tracking overview, oval trajectory at $k_t = 0.4$.

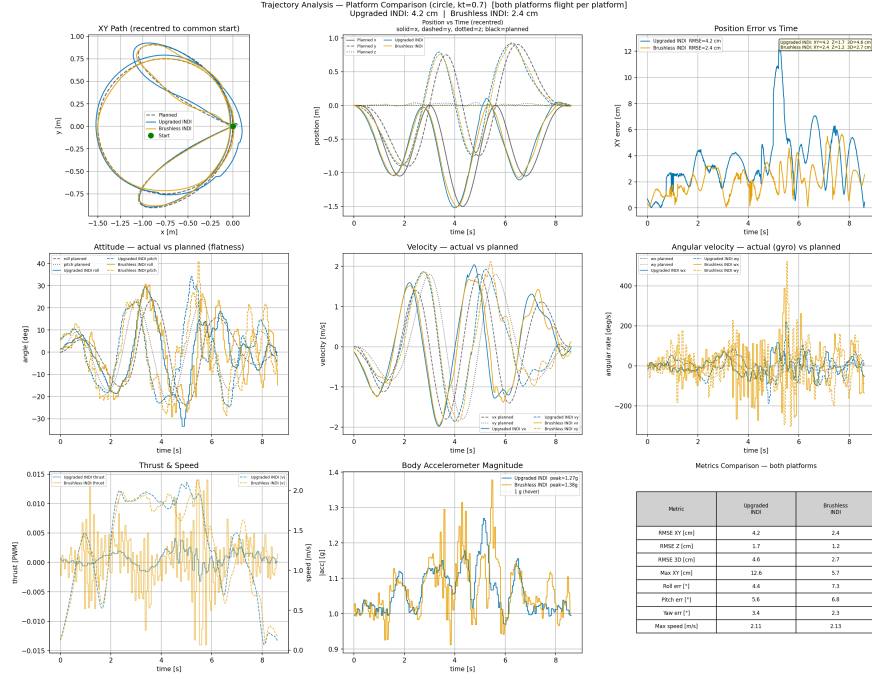


Figure 8: Upgraded vs. brushless tracking overview, circle trajectory at $k_t = 0.7$.

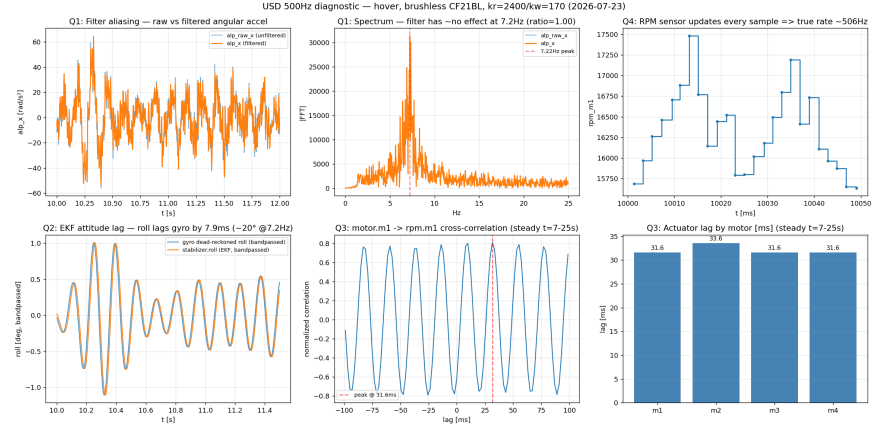


Figure 9: 500 Hz SD-card diagnostic at hover, brushless platform: the ≈ 7.22 Hz resonance quantified.

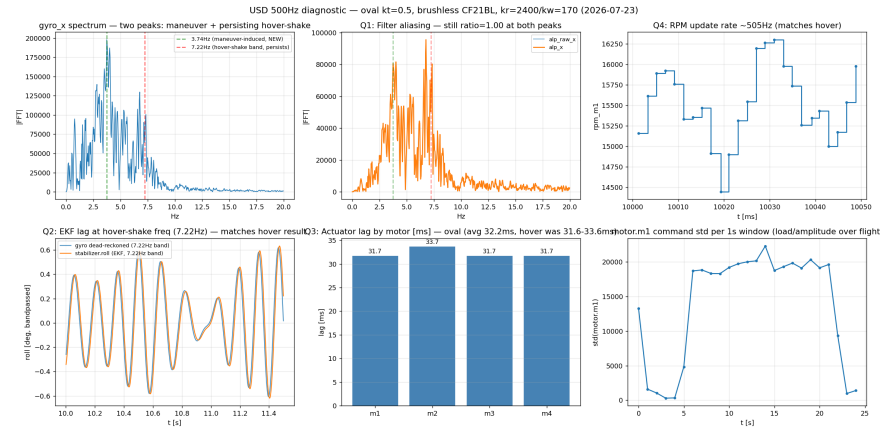


Figure 10: 500 Hz SD-card diagnostic under an oval maneuver, brushless platform: the resonance persists unchanged under real maneuvering load.

D Full k_t -Sweep Data

Full per- k_t data underlying the oval and circle summaries in Section 5.3, both platforms. “Crash” entries are excluded from the RMSE/error/speed columns for that k_t .

k_t	XY / Z RMSE		Roll/Pitch/Yaw err		Max speed [m/s]	
	Upgraded	Brushless	Upgraded	Brushless	Upgraded	Brushless
0.05	2.1/0.8 cm	2.3/0.5 cm	1.7/1.3/0.8°	2.2/2.9/0.9°	1.10	1.22
0.1	2.8/0.9 cm	1.3/0.7 cm	2.3/1.5/0.7°	3.1/1.6/0.3°	1.33	1.32
0.2	2.9/1.3 cm	2.5/1.1 cm	3.3/1.5/1.0°	2.6/1.3/0.4°	1.61	1.51
0.3	3.0/1.8 cm	2.8/1.0 cm	3.2/1.7/1.4°	2.9/1.5/0.5°	1.76	1.72
0.4	3.2/2.1 cm	3.2/1.4 cm	4.4/2.1/1.9°	10.5/6.1/2.0°	1.97	2.25
0.5	crash (5/5)	3.1/1.6 cm	—	5.5/2.7/1.4°	—	2.20
0.6	crash	not flown	—	—	—	—
0.7	not flown	crash	—	—	—	—

Table 6: Oval k_t sweep, both platforms.

k_t	XY / Z RMSE		Roll/Pitch/Yaw err		Max speed [m/s]	
	Upgraded	Brushless	Upgraded	Brushless	Upgraded	Brushless
0.05	1.7/0.8 cm	1.9/0.6 cm	2.1/1.8/0.8°	1.9/1.4/0.6°	1.03	1.21
0.1	1.9/1.3 cm	1.7/0.6 cm	1.9/1.8/1.1°	4.2/3.9/0.7°	1.15	1.50
0.2	2.3/1.2 cm	6.1/1.8 cm	2.2/1.8/1.4°	14.5/13.2/3.6°	1.35	1.71
0.3	2.2/1.5 cm	1.4/0.8 cm	2.3/2.1/1.9°	3.1/2.7/0.6°	1.53	1.75
0.4	2.7/1.8 cm	4.2/1.4 cm	3.5/3.6/2.5°	10.0/9.3/2.4°	1.73	1.84
0.5	2.8/1.6 cm	3.7/1.2 cm	3.5/3.5/2.6°	7.9/7.5/2.0°	1.84	1.88
0.6	3.7/2.4 cm	1.9/1.0 cm	3.1/3.3/2.7°	4.7/4.9/1.5°	1.98	1.84
0.7	4.2/1.7 cm	2.4/1.2 cm	4.4/5.6/3.4°	7.3/6.8/2.3°	2.11	2.13
0.8	3.3/2.5 cm	crash	4.8/5.3/3.6°	—	2.37	—
0.9	crash	crash	—	—	—	—
1.0	3.0/3.6 cm	not flown	4.9/4.0/4.6°	—	2.48	—

Table 7: Circle k_t sweep, both platforms.